Software Testing Documentation

Testing Framework & Objective-Aligned Results

This document contains the functional and validation testing results for the Dream Achievers Learning Center (DALC) System, structured strictly according to the 15 system modules and objectives specified in the Requirements Analysis Document (RAD). To ensure complete transparency and evaluative clarity for the defense panel, **each specific sub-module objective is presented in an independent, separated section** accompanied by its dedicated Functional Test Table and Validation Test Table.

MODULE 1: DASHBOARD MODULE

Sub-Module 1.1: Centralized Utility Interface

Specific Objective: This feature provides distinct, intuitive dashboard landing pages tailored specifically for the workflows of Admins, Teachers, and Guardians. The Admin dashboard includes system-wide metrics, while the Teacher dashboard focuses purely on classroom metrics. The Guardian portal utilizes enlarged utility icons for improved accessibility.

Table 1.1A. Functional Test result for Centralized Utility Interface

Centralized Utility Interface - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Centralized Utility Interface	Verify distinct dashboard landing pages tailored specifically for the workflows of Admins (system-wide metrics), Teachers (classroom metrics), and Guardians (enlarged utility icons for accessibility).	Admin, Teacher, and Guardian user accounts exist in the database with corresponding role bindings.	1. Authenticate with Administrator credentials and inspect /admin/dashboard. 2. Authenticate with Teacher credentials and inspect /teacher/dashboard. 3. Authenticate with Guardian credentials and inspect /guardian/dashboard.	Admin dashboard displays institutional KPI metrics (total students, active teachers, pending approvals, academic quarter status). Teacher dashboard displays classroom metrics (assigned roster count, active IEPs, daily log counters). Guardian portal renders WCAG-compliant high-contrast enlarged utility tiles.	**Pass**

Table 1.1B. Validation Test result for Centralized Utility Interface

Centralized Utility Interface - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Centralized Utility Interface	Role-Based Dashboard Navigation	User logs into the system with their specific role credentials (Admin, Teacher, or Guardian).	System displays the correct landing page tailored strictly to the user's workflow.	System successfully validated role credentials and displayed the tailored dashboard with appropriate metrics and accessibility icons.	**Working**

MODULE 2: USER MANAGEMENT AND SECURITY MODULE

Sub-Module 2.1: Account Generation

Specific Objective: This function allows the Administrator to systematically generate login credentials for Teachers and Guardians.

Table 2.1A. Functional Test result for Account Generation

Account Generation - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Account Generation	Verify systematic generation of login credentials for Teachers and Guardians with default role assignment and temporary credentials.	Administrator is authenticated and on /admin/accounts.	1. Click 'Add New Account'. 2. Select role ('Teacher' or 'Guardian'). 3. Enter user full name, email, and contact number. 4. Submit account generation form.	User record is created in the database, assigned designated role, and issued secure initial login credentials.	**Pass**

Table 2.1B. Validation Test result for Account Generation

Account Generation - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Account Generation	Account Creation Form Submission	Administrator submits new account details with required fields and email formatting.	System validates uniqueness of email and enforces valid phone/name formats.	System successfully validated account input parameters and systematically generated credentials with default role assignment.	**Working**

Sub-Module 2.2: Credential Management

Specific Objective: This feature provides the Administrator with the capability to manually update specific user account details, including modifying usernames, updating registered email addresses, and executing forced password resets.

Table 2.2A. Functional Test result for Credential Management

Credential Management - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Credential Management	Verify Administrator capability to manually update account details (username, email address) and execute forced password resets.	Administrator is authenticated; target user account exists in database.	1. Locate user in User Management table. 2. Click 'Edit Account', modify username and email address, and click 'Save Changes'. 3. Click 'Force Password Reset' on user action menu.	User record updates immediately in database; user password is reset to temporary value and flagged for mandatory change upon next authentication.	**Pass**

Table 2.2B. Validation Test result for Credential Management

Credential Management - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Credential Management	Account Detail Update & Forced Reset Trigger	Administrator edits account username/email and triggers a forced password reset.	System persists modified credentials and flags the user record for forced password change.	System successfully updated user account details and enforced password reset flag, invalidating prior sessions.	**Working**

Sub-Module 2.3: Authentication

Specific Objective: This security function enforces secure login mechanisms, including a mandatory password change upon the user's first login. It also implements Two-Factor Authentication (2FA) prior to executing critical administrative actions.

Table 2.3A. Functional Test result for Authentication

Authentication - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Authentication	Verify mandatory password change upon user's first login and enforcement of Two-Factor Authentication (2FA) prior to executing critical administrative actions.	New user with unreset initial credentials exists; Administrator with 2FA configured exists.	1. Log in with newly generated account using temporary password. 2. Verify immediate redirection to mandatory password change screen (/password/setup). 3. Log in as Administrator, navigate to critical administrative action, and initiate action. 4. Enter required TOTP 2FA code.	First-time user cannot navigate past /password/setup until password is changed; critical administrative action is blocked until valid 2FA code is confirmed.	**Pass**

Table 2.3B. Validation Test result for Authentication

Authentication - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Authentication	First-Time Login & Critical Administrative Action	User logs in with temporary credentials; Administrator executes protected action.	System restricts unauthorized navigation until password is updated and validates 6-digit TOTP code.	Mandatory password change was strictly enforced on first login, and 2FA authentication verified prior to critical action execution.	**Working**

Sub-Module 2.4: Role-Based Access Control (RBAC)

Specific Objective: This feature automatically assigns default user roles upon account generation, binding users to their predefined permission settings.

Table 2.4A. Functional Test result for Role-Based Access Control (RBAC)

Role-Based Access Control (RBAC) - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Role-Based Access Control (RBAC)	Verify automatic assignment of default user roles upon account generation, binding users strictly to their predefined permission settings.	Accounts exist with roles admin, teacher, and guardian.	1. Log in as teacher and attempt direct URL navigation to /admin/accounts. 2. Log in as guardian and attempt direct URL navigation to /teacher/daily-logs. 3. Inspect user model permissions in database.	HTTP 403 Forbidden is returned for cross-role URL navigation; users are strictly restricted to authorized role routes.	**Pass**

Table 2.4B. Validation Test result for Role-Based Access Control (RBAC)

Role-Based Access Control (RBAC) - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Role-Based Access Control (RBAC)	Unauthorized Route Access Attempt	User attempts to navigate to an unauthorized module URL outside assigned role.	System middleware validates role token against route permissions.	System successfully blocked unauthorized route access and bound users strictly to predefined role permissions.	**Working**

Sub-Module 2.5: User-Initiated Credential Recovery

Specific Objective: This function provides a forgot password feature on the login interface, allowing users to autonomously recover their accounts via secure email links.

Table 2.5A. Functional Test result for User-Initiated Credential Recovery

User-Initiated Credential Recovery - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
User-Initiated Credential Recovery	Verify forgot password feature on login interface, allowing autonomous account recovery via secure email links.	Registered user account exists with a verified email address.	1. Navigate to /login and click 'Forgot your password?'. 2. Enter registered email address and submit form. 3. Open password reset email and click secure recovery link. 4. Input new password, confirm, and submit.	Password reset email dispatched; link directs to valid reset token screen; user logs in successfully with new password.	**Pass**

Table 2.5B. Validation Test result for User-Initiated Credential Recovery

User-Initiated Credential Recovery - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
User-Initiated Credential Recovery	Password Reset Link Request & Verification	User initiates credential recovery from the login screen using registered email.	System generates cryptographic, time-limited token and validates reset form submission.	System successfully generated secure recovery token, dispatched email link, and updated credentials upon verification.	**Working**

Sub-Module 2.6: Teacher Privilege Management

Specific Objective: This feature allows the Administrator to manually grant additional administrative access levels to individual Teacher accounts.

Table 2.6A. Functional Test result for Teacher Privilege Management

Teacher Privilege Management - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Teacher Privilege Management	Verify Administrator capability to manually grant additional administrative access levels to individual Teacher accounts.	Administrator is authenticated; target Teacher account exists.	1. Navigate to Teacher Privilege Management section in User Management. 2. Select specific Teacher account. 3. Toggle specific administrative permissions (e.g., manage_guardian_accounts). 4. Save permission assignments.	Permissions persist in database; Teacher account receives elevated access permissions.	**Pass**

Table 2.6B. Validation Test result for Teacher Privilege Management

Teacher Privilege Management - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Teacher Privilege Management	Teacher Permission Elevation Update	Administrator grants administrative access tile permissions to a Teacher account.	System stores granular permission flags in database and binds them to the Teacher session.	System successfully granted elevated permissions to target Teacher without elevating the base role.	**Working**

MODULE 3: ADMIN ACCESS MODULE

Sub-Module 3.1: Granted Admin Workspace

Specific Objective: A dynamic dashboard section that displays specific administrative access tiles based on the Teacher's assigned permissions.

Table 3.1A. Functional Test result for Granted Admin Workspace

Granted Admin Workspace - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Granted Admin Workspace	Verify dynamic dashboard section displaying specific administrative access tiles based on the Teacher's assigned permissions.	Teacher user has been granted specific administrative privileges by Administrator.	1. Log in as Teacher with deputized permissions. 2. View Teacher dashboard landing page. 3. Verify appearance of 'Granted Admin Workspace' section and corresponding access tiles.	Dynamic workspace section renders displaying tiles strictly corresponding to granted permissions.	**Pass**

Table 3.1B. Validation Test result for Granted Admin Workspace

Granted Admin Workspace - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Granted Admin Workspace	Deputized Teacher Dashboard Loading	Teacher with assigned administrative privileges accesses personal dashboard.	System dynamically checks permission records and renders corresponding administrative tiles.	System successfully rendered dynamic administrative access tiles based strictly on granted Teacher permissions.	**Working**

Sub-Module 3.2: Authorized Task Execution

Specific Objective: Allows designated Teachers to perform assigned administrative functions directly from their standard accounts via the granted modules.

Table 3.2A. Functional Test result for Authorized Task Execution

Authorized Task Execution - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Authorized Task Execution	Verify designated Teachers can perform assigned administrative functions directly from their standard accounts via the granted modules.	Teacher has granted permission to manage Guardian accounts.	1. Log in as designated Teacher. 2. Click on the granted Guardian Management tile. 3. Execute administrative action (e.g., update guardian contact details or create new guardian). 4. Save the transaction.	Action executes successfully without requiring switching to an administrator account.	**Pass**

Table 3.2B. Validation Test result for Authorized Task Execution

Authorized Task Execution - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Authorized Task Execution	Administrative Function Execution via Teacher Account	Deputized Teacher executes granted administrative tasks from standard account.	System validates deputized permission token and commits the administrative transaction.	System successfully executed authorized administrative action directly from the designated Teacher account.	**Working**

Sub-Module 3.3: Scoped Credential Management

Specific Objective: A security restriction that limits deputized Teachers to managing only Guardian credentials, automatically blocking access to Teacher or Administrator accounts.

Table 3.3A. Functional Test result for Scoped Credential Management

Scoped Credential Management - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Scoped Credential Management	Verify security restriction limiting deputized Teachers to managing only Guardian credentials, automatically blocking access to Teacher or Administrator accounts.	Teacher has granted account management permissions.	1. Log in as deputized Teacher and open User Management interface. 2. Inspect user list filter and action buttons. 3. Attempt to view, edit, or reset credentials of another Teacher or Administrator.	Only Guardian accounts are visible and editable; Teacher and Administrator accounts are hidden and inaccessible.	**Pass**

Table 3.3B. Validation Test result for Scoped Credential Management

Scoped Credential Management - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Scoped Credential Management	Cross-Role Credential Management Attempt	Deputized Teacher attempts to access or modify Teacher or Admin credentials.	System applies scope filter on account queries and controller authorization checks.	System strictly blocked access to Teacher/Admin accounts and restricted scope exclusively to Guardian credentials.	**Working**

MODULE 4: STUDENT PROFILING MODULE

Sub-Module 4.1: Initial Profile Creation

Specific Objective: This function allows Administrators to execute the initial creation of a student profile. This includes inputting basic demographic information, diagnosis, document files, and systematically linking the student's profile to their respective generated Guardian account.

Table 4.1A. Functional Test result for Initial Profile Creation

Initial Profile Creation - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Initial Profile Creation	Verify Administrator initial creation of a student profile, including inputting demographic info, diagnosis, document files, and systematically linking to a generated Guardian account.	Administrator is authenticated; registered Guardian account is available for linkage.	1. Navigate to /admin/students/create. 2. Enter demographic data (Name, LRN, DOB, Gender, Address). 3. Input developmental diagnosis and upload document files (medical assessment PDF). 4. Select and link corresponding Guardian account. 5. Submit form.	Student profile is created, attached documents stored in secure vault, and foreign key linkage to Guardian account established.	**Pass**

Table 4.1B. Validation Test result for Initial Profile Creation

Initial Profile Creation - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Initial Profile Creation	Student Profile Creation Form Submission	Administrator inputs demographic details, attaches assessment documents, and links Guardian account.	System validates mandatory fields, LRN format, file mime types, and foreign key relations.	System successfully created student profile, securely stored documents, and linked profile to Guardian account.	**Working**

Sub-Module 4.2: Profile Deletion Guard

Specific Objective: This security feature enforces a strict confirmation protocol, requiring the user to manually type the word 'confirm' before executing the deletion of a student profile.

Table 4.2A. Functional Test result for Profile Deletion Guard

Profile Deletion Guard - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Profile Deletion Guard	Verify security protocol requiring the user to manually type the exact word 'confirm' before executing deletion of a student profile.	Existing student profile selected for deletion by Administrator.	1. Click 'Delete Student Profile' button on student view. 2. In confirmation modal, type 'confirm' into verification input field. 3. Click final delete button.	Profile deletion proceeds only when the exact confirmation string is provided.	**Pass**

Table 4.2B. Validation Test result for Profile Deletion Guard

Profile Deletion Guard - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**

Profile Deletion Guard	Student Profile Deletion Request	User initiates deletion and enters confirmation string in security modal.	System verifies that entered text matches 'confirm' exactly before dispatching DELETE request.	System successfully prevented accidental deletion and executed profile removal only upon exact 'confirm' validation.	**Working**

Sub-Module 4.3: Enrollment Logic

Specific Objective: This workflow function strictly separates enrollment actions; 'Enroll Student' opens a blank form for brand new profiles, whereas 'Re-Enroll Student' opens a specific interface to populate data from existing or frozen student records.

Table 4.3A. Functional Test result for Enrollment Logic

Enrollment Logic - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Enrollment Logic	Verify strict separation of enrollment actions: 'Enroll Student' opens blank form for brand new profiles; 'Re-Enroll Student' opens specific interface populating data from existing or frozen student records.	Administrator is on Student Management page; both new applicant and archived/inactive student records exist.	1. Click 'Enroll Student' button and inspect form fields. 2. Return to dashboard and click 're-Enroll Student'. 3. Select an existing/frozen student record from selection list.	'Enroll Student' provides clean, empty input fields; 'Re-Enroll Student' pre-populates existing historical profile and demographic data for review and reactivation.	**Pass**

Table 4.3B. Validation Test result for Enrollment Logic

Enrollment Logic - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Enrollment Logic	Enrollment vs Re-Enrollment Action Trigger	Administrator initiates brand new enrollment and historical student re-enrollment.	System routes 'Enroll Student' to blank state and 'Re-Enroll Student' to record selection and hydration.	System strictly separated enrollment workflows, providing blank forms for new learners and pre-populated data for re-enrolling learners.	**Working**

MODULE 5: DAILY LOGGING MODULE

Sub-Module 5.1: Input Daily Notes

Specific Objective: This feature provides Teachers with a dedicated interface to record raw, day-to-day behavioral and academic observations for their assigned students.

Table 5.1A. Functional Test result for Input Daily Notes

Input Daily Notes - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Input Daily Notes	Verify dedicated interface for Teachers to record raw, day-to-day behavioral and academic observations for assigned students.	Teacher is logged in and assigned to student roster.	1. Navigate to /teacher/daily-logs/create. 2. Select assigned student and input date. 3. Type raw observations into academic and behavioral observation fields. 4. Save daily notes.	Raw notes are stored in database associated with student, date, and authoring teacher.	**Pass**

Table 5.1B. Validation Test result for Input Daily Notes

Input Daily Notes - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Input Daily Notes	Daily Note Entry Submission	Teacher inputs classroom behavioral and academic notes for an assigned student.	System validates note presence, date constraints, and teacher-student roster linkage.	System successfully accepted and persisted raw observational notes under authoring teacher and assigned student.	**Working**

Sub-Module 5.2: Formal Documentation Generation

Specific Objective: This AI-powered editing function takes a teacher's quick, typed classroom notes and utilizes Artificial Intelligence to rewrite them into a formal, structured daily log. The AI-generated text is saved directly under the 'Daily Logs' header. Once finalized, this documentation can be exported and downloaded as a PDF document.

Table 5.2A. Functional Test result for Formal Documentation Generation

Formal Documentation Generation - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Formal Documentation Generation	Verify AI-powered editing function that takes quick, typed classroom notes, rewrites them into a formal structured daily log under 'Daily Logs', and exports/downloads as a PDF document.	Raw daily notes entered for a student.	1. Click 'Generate Formal Log (AI)' button on daily notes form. 2. Inspect generated text placed under 'Daily Logs' header. 3. Review, make minor edits, and click 'Finalize & Save'. 4. Click 'Download PDF' button.	AI structures raw notes into professional clinical prose; formatted log is saved; clean PDF document is compiled and downloaded.	**Pass**

Table 5.2B. Validation Test result for Formal Documentation Generation

Formal Documentation Generation - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Formal Documentation Generation	AI Formal Log Generation & PDF Export	Teacher triggers AI rewrite on raw notes and exports the finalized documentation.	System processes raw notes through AI agent, places text under 'Daily Logs', and compiles PDF.	System successfully generated formal clinical prose from notes and compiled downloadable PDF document.	**Working**

MODULE 6: REPORT MANAGEMENT AND GENERATION MODULE

Sub-Module 6.1: Report List and Filtering

Specific Objective: This interface serves as the primary gateway for teacher reports. It provides a search and filter system to manage high data volumes, allowing Teachers to select specific students, view historical data, and preview approved documents.

Table 6.1A. Functional Test result for Report List and Filtering

Report List and Filtering - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Report List and Filtering	Verify primary gateway for teacher reports with search and filter system by student, report type, school year, and preview approved documents.	Teacher is authenticated; historical reports exist in database.	1. Navigate to /teacher/reports. 2. Filter list by student name and report type. 3. Click 'Preview' on an approved report.	Report table filters dynamically; clicking preview renders modal/inline view of approved document.	**Pass**

Table 6.1B. Validation Test result for Report List and Filtering

Report List and Filtering - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Report List and Filtering	Report Search and Filter Query	Teacher searches and filters report archive by student name, year, and quarter.	System queries indexed report tables and returns scoped matching records.	System successfully filtered high volume of report records and rendered instant document previews.	**Working**

Sub-Module 6.2: Integrated Report List

Specific Objective: Provides Guardians with a chronological list of their child's approved reports to securely preview and download PDF copies of Quarterly, Mid-Year, and Year-End Reports.

Table 6.2A. Functional Test result for Integrated Report List

Integrated Report List - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Integrated Report List	Verify chronological list of approved reports for Guardians to securely preview and download PDF copies of Quarterly, Mid-Year, and Year-End Reports.	Guardian is authenticated and linked to a student with approved reports.	1. Log in as Guardian and navigate to 'Child Reports' tab. 2. Inspect chronological order. 3. Click 'Preview' and 'Download PDF'.	Approved reports display in chronological order; preview displays report content; PDF download streams securely.	**Pass**

Table 6.2B. Validation Test result for Integrated Report List

Integrated Report List - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Integrated Report List	Guardian Report List Retrieval	Guardian accesses student report repository and downloads approved PDF summaries.	System validates Guardian-child association and conceals draft/unapproved reports.	System successfully delivered chronological list of approved reports and streamed authentic PDF files.	**Working**

Sub-Module 6.3: Generate New Report

Specific Objective: Accessed directly from within the Report List, this function allows Teachers to trigger the creation of a new document.

Table 6.3A. Functional Test result for Generate New Report

Generate New Report - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Generate New Report	Verify Teacher can trigger creation of a new report directly from within Report List interface.	Teacher is on Report List interface.	1. Click '+ Generate New Report' button. 2. Select target student, academic year, and report period. 3. Proceed to report builder.	Interface initializes a new report builder session with selected student metadata pre-loaded.	**Pass**

Table 6.3B. Validation Test result for Generate New Report

Generate New Report - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Generate New Report	New Report Generation Trigger	Teacher selects student and report period from the Report List to initialize document.	System creates new report entity in draft status linked to student and grading period.	System successfully initialized new report workspace and loaded corresponding learner metadata.	**Working**

Sub-Module 6.4: Generate Draft

Specific Objective: This AI integration feature generates drafts for every report type, requiring manual Teacher review prior to submission.

Table 6.4A. Functional Test result for Generate Draft

Generate Draft - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Generate Draft	Verify AI integration generates comprehensive drafts for every report type, requiring manual Teacher review prior to submission.	Report builder open; underlying student daily logs and IEP milestones exist.	1. Click 'Generate Draft (AI)' button in report builder. 2. Inspect AI-generated summary content. 3. Make manual edits and click 'Save Draft'.	AI compiles structured draft; content is placed in editable fields; manual edits are saved; status remains 'Draft' until submitted.	**Pass**

Table 6.4B. Validation Test result for Generate Draft

Generate Draft - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Generate Draft	AI Draft Generation Request	Teacher requests AI draft generation across selected report period.	System synthesizes student logs, produces structured draft, and retains draft status pending review.	System successfully generated cohesive draft and enforced mandatory teacher review prior to submission.	**Working**

Sub-Module 6.5: Automated Report Stamping

Specific Objective: This function systematically appends the School Administrator's authorized digital stamp to all finalized and approved reports.

Table 6.5A. Functional Test result for Automated Report Stamping

Automated Report Stamping - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**

Automated Report Stamping	Verify systematic appending of School Administrator's authorized digital stamp to all finalized and approved reports.	Report has been reviewed and approved by Administrator; digital stamp configured.	1. Administrator approves pending report. 2. Generate/view finalized PDF copy. 3. Inspect signature/stamp footer section.	Official administrative digital stamp and verification timestamp are systematically rendered on finalized PDF.	**Pass**

Table 6.5B. Validation Test result for Automated Report Stamping

Automated Report Stamping - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Automated Report Stamping	Report Approval & PDF Compilation	Administrator marks report approved and user opens finalized PDF.	System embeds authorized digital stamp image and verification metadata onto PDF canvas.	System successfully appended official digital stamp exclusively to finalized and approved reports.	**Working**

Sub-Module 6.6: Admin Approval Workflow

Specific Objective: This function allows administrators to approve major academic summaries or reject them using a 'Revision Notes' tab to send specific feedback back to the teacher for necessary corrections.

Table 6.6A. Functional Test result for Admin Approval Workflow

Admin Approval Workflow - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Admin Approval Workflow	Verify Administrator workflow to approve academic summaries or reject them using a 'Revision Notes' tab to send specific feedback to the teacher.	Teacher has submitted report for review (status pending_approval).	1. Log in as Administrator and navigate to Report Approval inbox. 2. Open submitted report, switch to 'Revision Notes' tab. 3. Enter feedback and click 'Reject / Request Revision'.	Report status transitions to rejected / revision_requested; revision notes are recorded; notification sent to teacher.	**Pass**

Table 6.6B. Validation Test result for Admin Approval Workflow

Admin Approval Workflow - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Admin Approval Workflow	Report Review Decision Submission	Administrator rejects a submitted summary with explanatory feedback notes.	System requires non-empty revision notes and transitions document status to revision_requested.	System successfully registered rejection feedback, updated report status, and notified authoring teacher.	**Working**

Sub-Module 6.7: Home-Based Recommendation Generator

Specific Objective: Integrated within the Daily Report, this AI-driven function analyzes the teacher's daily logs to automatically generate practical learning tips and home-based activities. The AI-generated recommendations serve as an editable draft, allowing the teacher to review, edit, and refine the content before finalizing it for the parents.

Table 6.7A. Functional Test result for Home-Based Recommendation Generator

Home-Based Recommendation Generator - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Home-Based Recommendation Generator	Verify AI-driven function analyzing daily logs to generate practical learning tips and home-based activities as an editable draft for teacher refinement.	Student has accumulated daily logs in current period.	1. Open Daily Report editor. 2. Click 'Generate Home Recommendations (AI)'. 3. Inspect generated tips; edit recommendations. 4. Save finalized Daily Report.	AI produces age-appropriate, actionable home activities; text remains fully editable; saved recommendations attach to daily report.	**Pass**

Table 6.7B. Validation Test result for Home-Based Recommendation Generator

Home-Based Recommendation Generator - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Home-Based Recommendation Generator	AI Home Recommendation Trigger	Teacher requests home-based recommendations for Daily Report.	System analyzes classroom observations and populates editable suggestion draft.	System successfully produced actionable home-based activities grounded in teacher observation logs.	**Working**

Sub-Module 6.8: Secure Media Attachment

Specific Objective: Available exclusively for Daily Reports, this feature allows Teachers to upload activity photos that are automatically compressed and resized for storage efficiency. To ensure privacy, all media files bypass public directories and are strictly restricted to authorized active sessions (Admin, assigned Teacher, and Guardian).

Table 6.8A. Functional Test result for Secure Media Attachment

Secure Media Attachment - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Secure Media Attachment	Verify activity photo upload exclusively for Daily Reports, automated compression/resizing, and storage bypassing public directories restricted to authorized sessions.	Teacher is authoring Daily Report with photo attachment.	1. Upload high-resolution photo in Daily Report editor. 2. Save report and inspect stored file dimensions/size. 3. Attempt direct HTTP URL access from unauthenticated browser.	Photo is automatically compressed and resized; saved in private storage directory; direct unauthenticated URL request returns 403 Forbidden.	**Pass**

Table 6.8B. Validation Test result for Secure Media Attachment

Secure Media Attachment - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Secure Media Attachment	Media Upload & Vault Access Request	Teacher uploads activity photo; unauthenticated user attempts direct file URL access.	System resizes image, stores in non-public vault, and enforces session authorization checks.	System successfully compressed media and restricted access strictly to authorized active sessions.	**Working**

Sub-Module 6.9: Daily Reports / Home Activities

Specific Objective: This Provides a dedicated interface for Guardians to view teacher-approved AI intervention strategies and activities to support their child's developmental goals at home alongside the Daily Report.

Table 6.9A. Functional Test result for Daily Reports / Home Activities

Daily Reports / Home Activities - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Daily Reports / Home Activities	Verify dedicated interface for Guardians to view teacher-approved AI intervention strategies and activities supporting child's developmental goals at home.	Teacher has finalized and approved Daily Report containing home recommendations.	1. Log in as Guardian and navigate to 'Home Activities & Daily Reports'. 2. Select report date. 3. View developmental summary and recommended home activities list.	Guardian interface cleanly renders approved home activities with visual goal indicators.	**Pass**

Table 6.9B. Validation Test result for Daily Reports / Home Activities

Daily Reports / Home Activities - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**

Daily Reports / Home Activities	Guardian Home Activities Portal Access	Guardian accesses Daily Reports / Home Activities portal for their child.	System fetches teacher-approved daily reports and renders recommended home interventions.	System successfully displayed teacher-approved intervention strategies and activities on the Guardian portal.	**Working**

Sub-Module 6.10: AI Report Revision

Specific Objective: Enables teachers to automatically rewrite rejected reports using an AI button based on the administrator's revision notes.

Table 6.10A. Functional Test result for AI Report Revision

AI Report Revision - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
AI Report Revision	Verify Teacher can automatically rewrite rejected reports using an AI button that incorporates Administrator revision notes into revised draft.	Report has status revision_requested with Administrator feedback in revision notes.	1. Teacher opens rejected report. 2. Review Administrator's revision notes. 3. Click 'Revise with AI' button. 4. Inspect rewritten text and resubmit report.	AI synthesizes original text and revision notes to produce updated narrative addressing all noted deficiencies; resubmits to pending_approval.	**Pass**

Table 6.10B. Validation Test result for AI Report Revision

AI Report Revision - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
AI Report Revision	AI Report Revision Execution	Teacher clicks AI revision button on a report with administrator feedback.	System sends original draft and revision notes to AI service to synthesize corrected narrative.	System successfully generated revised report addressing administrative notes for rapid resubmission.	**Working**

MODULE 7: ARCHIVING MODULE

Sub-Module 7.1: Year-End Rollover (Long-Term Record Archiving)

Specific Objective: A dedicated administrative function for bulk-archiving students. Initiating the 'Run Archive' process requires a mandatory Two-Factor Authentication (2FA) code. This generates a permanent Snapshot of the student information, IEP, Quarterly Summaries, Mid-Year Summary, and Year-End Summary, while simultaneously freezing all associated student files into a secure, read-only state and clearing the active 'Daily Logs' of all teachers to reset the workspace for the upcoming school year.

Table 7.1A. Functional Test result for Year-End Rollover (Long-Term Record Archiving)

Year-End Rollover (Long-Term Record Archiving) - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Year-End Rollover (Long-Term Record Archiving)	Verify bulk-archiving of students requiring mandatory 2FA code, creating permanent snapshots (info, IEP, summaries), freezing student files read-only, and clearing active Daily Logs for upcoming school year.	End of academic year reached; Administrator has 2FA enabled.	1. Navigate to /admin/archive/rollover. 2. Click 'Run Archive' and enter 2FA verification code. 3. Confirm archive execution. 4. Inspect database snapshots, student file status, and active teacher daily logs table.	Snapshots generated; associated student files frozen to read-only; active daily logs cleared; new school year reset.	**Pass**

Table 7.1B. Validation Test result for Year-End Rollover (Long-Term Record Archiving)

Year-End Rollover (Long-Term Record Archiving) - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Year-End Rollover (Long-Term Record Archiving)	Year-End Archive Execution Request	Administrator initiates bulk archive with 2FA authentication.	System validates 2FA token, snapshots academic records, sets files to read-only, and resets logs.	System successfully generated permanent snapshots, froze records to read-only, and reset active daily logs.	**Working**

Sub-Module 7.2: Quarterly Rollover (Daily Report Cleanup)

Specific Objective: This automated background feature deletes generated Daily Reports at the end of each grading period to ensure active dashboards remain uncluttered for Teachers and Guardians. The system safely executes this purge to optimize storage while preserving the teacher's Daily Logs throughout the active school year so they remain available for generating Mid-Year and Year-End Summaries prior to the Year-End Rollover.

Table 7.2A. Functional Test result for Quarterly Rollover (Daily Report Cleanup)

Quarterly Rollover (Daily Report Cleanup) - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Quarterly Rollover (Daily Report Cleanup)	Verify automated background deletion of generated Daily Reports at end of grading period to unclutter dashboards while strictly preserving teacher Daily Logs throughout active school year.	Grading quarter reaches end date; Daily Reports exist alongside underlying Daily Logs.	1. Trigger quarterly rollover maintenance command or reach scheduled quarter boundary. 2. Inspect Daily Reports dashboard table. 3. Inspect underlying Daily Logs database table.	Generated Daily Reports are cleaned up from active dashboard; raw Daily Logs remain intact for Mid-Year and Year-End summary generation.	**Pass**

Table 7.2B. Validation Test result for Quarterly Rollover (Daily Report Cleanup)

Quarterly Rollover (Daily Report Cleanup) - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Quarterly Rollover (Daily Report Cleanup)	Quarterly Purge Background Job Trigger	System executes quarterly cleanup at the close of an academic quarter.	System purges generated daily reports while preserving foundational daily observation logs.	System safely cleared active daily reports from dashboard view while preserving raw daily logs intact.	**Working**

MODULE 8: IEP AND GOAL MANAGEMENT MODULE

Sub-Module 8.1: IEP Management

Specific Objective: This feature enables teachers to build custom IEPs by selecting a student and development domain, then inputting their Present Level of Performance (PLOP). Teachers can generate AI-suggested learning objectives and activities, while retaining complete control to edit the AI suggestions or manually add their own before saving.

Table 8.1A. Functional Test result for IEP Management

IEP Management - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
IEP Management	Verify Teacher can build custom IEPs by selecting student and development domain, inputting PLOP, generating AI-suggested learning objectives and activities, with manual editing before saving.	Teacher is authenticated; assigned student exists.	1. Navigate to /teacher/iep/create. 2. Select student and development domain (Cognitive, Motor, Communication). 3. Input Present Level of Performance (PLOP). 4. Click 'Suggest Objectives & Activities (AI)'. 5. Edit suggested text or add custom objectives and save IEP.	AI suggestions populate fields; teacher edits persist; custom IEP record is saved.	**Pass**

Table 8.1B. Validation Test result for IEP Management

IEP Management - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
IEP Management	Custom IEP Construction Submission	Teacher builds IEP with domain selection, PLOP baseline, and AI objectives.	System validates domain data, incorporates teacher edits, and stores structured IEP document.	System successfully generated AI suggestions, allowed custom teacher editing, and saved finalized IEP.	**Working**

Sub-Module 8.2: Goal Management

Specific Objective: This interface allows Teachers to add, edit, delete, or comment on specific goals.

Table 8.2A. Functional Test result for Goal Management

Goal Management - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Goal Management	Verify Teacher interface to add, edit, delete, or comment on specific individual goals within an IEP.	Existing IEP loaded in Teacher portal.	1. Open student IEP and click 'Add Goal'. 2. Enter goal description, target completion date, and measurement criteria. 3. Edit an existing goal's description. 4. Post a comment on the goal. 5. Delete an obsolete goal.	Goal is added, modified, commented on, and deleted respectively with real-time UI updates.	**Pass**

Table 8.2B. Validation Test result for Goal Management

Goal Management - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**

Goal Management	Goal Entity CRUD Operations	Teacher performs addition, modification, comment attachment, and deletion on IEP goals.	System updates goal records and preserves referential integrity with parent IEP.	System successfully executed goal creation, updates, threaded comments, and deletion operations.	**Working**

Sub-Module 8.3: Progress and Goal Tracking

Specific Objective: Allows Guardians to independently monitor their child's current learning objectives and real-time progress statuses (e.g., In Progress, Achieved) established in the IEP.

Table 8.3A. Functional Test result for Progress and Goal Tracking

Progress and Goal Tracking - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Progress and Goal Tracking	Verify Guardians can independently monitor their child's current learning objectives and real-time progress statuses (e.g., In Progress, Achieved) established in the IEP.	Guardian is logged in and child has active IEP goals with updated progress statuses.	1. Log in as Guardian and navigate to 'IEP Progress & Goals' tab. 2. View list of domains and current learning objectives. 3. Inspect progress badges ('In Progress', 'Achieved') and percentage completion bars.	Learning objectives display accurately with corresponding real-time status indicators and progress metrics.	**Pass**

Table 8.3B. Validation Test result for Progress and Goal Tracking

Progress and Goal Tracking - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Progress and Goal Tracking	Guardian IEP Progress Inspection	Guardian accesses IEP tracking tab to monitor real-time milestone achievement.	System queries active IEP goals and renders real-time status badges and progress meters.	System successfully rendered accurate, real-time learning objective progress statuses for Guardian monitoring.	**Working**

MODULE 9: ANNOUNCEMENT MODULE

Sub-Module 9.1: Broadcast School Notices

Specific Objective: This function allows Administrators to create, publish, and subsequently edit official school notices, which are directly viewable by Teachers and Guardians on their respective dashboards.

Table 9.1A. Functional Test result for Broadcast School Notices

Broadcast School Notices - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Broadcast School Notices	Verify Administrator can create, publish, and subsequently edit official school notices directly viewable by Teachers and Guardians on their dashboards.	Administrator is authenticated; Teachers and Guardians are active.	1. Navigate to /admin/announcements. 2. Click 'Create Notice', input title, announcement body, target audience (All, Teachers, Guardians), and click 'Publish'. 3. Edit published notice text and save changes. 4. Log in as Teacher and Guardian to inspect dashboard notices.	Notice publishes and updates successfully; renders prominently on Teacher and Guardian dashboards.	**Pass**

Table 9.1B. Validation Test result for Broadcast School Notices

Broadcast School Notices - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Broadcast School Notices	School Notice Publishing & Modification	Administrator publishes official bulletin and updates notice contents.	System validates permissions, commits notice to database, and broadcasts to audience dashboards.	System successfully published and edited official notices, displaying them prominently on user dashboards.	**Working**

MODULE 10: COLLABORATIVE FEEDBACK MODULE

Sub-Module 10.1: Collaborative Feedback Thread

Specific Objective: Serving as the central communication hub, this feature allows Guardians, Teachers, and Administrators to view, leave, and reply to threaded comments tied directly to specific topics, such as individual student reports or specific IEP goals.

Table 10.1A. Functional Test result for Collaborative Feedback Thread

Collaborative Feedback Thread - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Collaborative Feedback Thread	Verify central communication hub allowing Guardians, Teachers, and Administrators to view, leave, and reply to threaded comments tied directly to specific topics (student reports or IEP goals).	Student report or IEP goal exists; all three roles have access to the student context.	1. Open specific student report as Guardian and post a comment in feedback thread. 2. Log in as Teacher, open same report, view Guardian's comment, and post a threaded reply. 3. Log in as Administrator and view complete threaded conversation.	Comments and replies appear in chronological threaded order with author tags and timestamps.	**Pass**

Table 10.1B. Validation Test result for Collaborative Feedback Thread

Collaborative Feedback Thread - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Collaborative Feedback Thread	Threaded Comment Submission & Association	Users across Admin, Teacher, and Guardian roles participate in topic-specific thread.	System links comments polymorphically to target report/goal and renders chronological hierarchy.	System successfully maintained threaded conversations tied directly to specific reports and IEP goals.	**Working**

MODULE 11: ARTIFICIAL INTELLIGENCE (AI) INTEGRATION MODULE

Sub-Module 11.1: AI-Powered Daily Log Generator

Specific Objective: This feature processes raw teacher observations and converts them into structured formal logs.

Table 11.1A. Functional Test result for AI-Powered Daily Log Generator

AI-Powered Daily Log Generator - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
AI-Powered Daily Log Generator	Verify processing of raw teacher observations and conversion into structured formal clinical logs.	Raw observation notes entered in Daily Log interface.	1. Input brief raw notes: 'John had difficulty with pencil grip during coloring; calmed down after sensory swing.' 2. Click 'AI Format Log'. 3. Inspect formatted output.	AI outputs formal prose categorized into Behavioral Observations, Fine Motor Skills, and Sensory Interventions.	**Pass**

Table 11.1B. Validation Test result for AI-Powered Daily Log Generator

AI-Powered Daily Log Generator - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
AI-Powered Daily Log Generator	Raw Note AI Conversion Trigger	System transmits informal teacher classroom observations to AI processing service.	System formats observations into clinical SPED structure and validates output safety.	System successfully converted unstructured observation notes into formal clinical daily logs.	**Working**

Sub-Module 11.2: AI IEP Goal Assistant

Specific Objective: This feature analyzes baseline assessments to suggest SMART-formatted learning goals and strategies.

Table 11.2A. Functional Test result for AI IEP Goal Assistant

AI IEP Goal Assistant - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
AI IEP Goal Assistant	Verify analysis of baseline assessments to suggest SMART-formatted learning goals and strategies.	PLOP baseline entered for a chosen developmental domain.	1. Input PLOP baseline: 'Student can recognize numbers 1-5 with 50% accuracy.' 2. Click 'Generate SMART Goal (AI)'. 3. Inspect output.	AI outputs specific, measurable, achievable, relevant, and time-bound goals with instructional strategies.	**Pass**

Table 11.2B. Validation Test result for AI IEP Goal Assistant

AI IEP Goal Assistant - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
AI IEP Goal Assistant	SMART Goal Generation Request	AI analyzes student baseline performance metrics to formulate SMART targets.	System enforces SMART criteria (condition, behavior, criterion, timeframe) on response.	System successfully analyzed baseline assessment and generated compliant SMART learning goals.	**Working**

Sub-Module 11.3: AI Progress Summary Generator

Specific Objective: This feature analyzes compiled daily logs to automatically draft cohesive summary paragraphs describing developmental trends.

Table 11.3A. Functional Test result for AI Progress Summary Generator

AI Progress Summary Generator - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
AI Progress Summary Generator	Verify analysis of compiled daily logs to automatically draft cohesive summary paragraphs describing developmental trends.	Student has multiple daily logs recorded across academic quarter.	1. Open Quarterly Summary report builder. 2. Click 'Generate Trend Summary (AI)'. 3. Inspect synthesized summary paragraphs.	AI synthesizes behavioral and academic trends across the quarter into cohesive, objective summary paragraphs.	**Pass**

Table 11.3B. Validation Test result for AI Progress Summary Generator

AI Progress Summary Generator - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
AI Progress Summary Generator	Quarterly Trend Analysis Synthesis	AI compiles historical daily logs to construct developmental trend summaries.	System filters logs by academic period and generates cohesive quarterly narrative.	System successfully synthesized multiple observation logs into professional progress summaries.	**Working**

Sub-Module 11.4: AI Home Recommendation Generator

Specific Objective: This feature analyzes daily logs to automatically suggest practical learning tips and home-based intervention strategies for Guardians.

Table 11.4A. Functional Test result for AI Home Recommendation Generator

AI Home Recommendation Generator - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
AI Home Recommendation Generator	Verify analysis of daily logs to automatically suggest practical learning tips and home-based intervention strategies for Guardians.	Daily log entry completed by Teacher.	1. Click 'Generate Home Recommendations' in Daily Report module. 2. Inspect generated activities and parent tips.	Practical, low-resource home activities and positive reinforcement tips generated based on today's classroom observations.	**Pass**

Table 11.4B. Validation Test result for AI Home Recommendation Generator

AI Home Recommendation Generator - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
AI Home Recommendation Generator	Parent Intervention Generation Trigger	AI processes classroom observations to suggest supportive home routines.	System extracts behavioral triggers and suggests practical parental reinforcement activities.	System successfully formulated actionable, supportive home learning tips based on classroom data.	**Working**

Sub-Module 11.5: AI Report Revision Assistant

Specific Objective: This feature analyzes the Administrator's rejection feedback alongside the teacher's original draft to automatically rewrite and correct rejected reports for faster resubmission.

Table 11.5A. Functional Test result for AI Report Revision Assistant

AI Report Revision Assistant - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**

AI Report Revision Assistant	Verify analysis of Administrator's rejection feedback alongside teacher's original draft to automatically rewrite and correct rejected reports for faster resubmission.	Report has been rejected with Administrator revision notes.	1. Open rejected report as Teacher. 2. Click 'AI Revision Assistant'. 3. Compare original text, feedback notes, and AI-revised draft.	AI specifically addresses and resolves issues highlighted in revision notes while preserving accurate historical observations.	**Pass**

Table 11.5B. Validation Test result for AI Report Revision Assistant

AI Report Revision Assistant - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
AI Report Revision Assistant	Feedback Correction AI Processing	AI analyzes administrator revision notes and reformulates draft text.	System integrates specific correction directives into revised narrative draft.	System successfully corrected draft reports based on administrator feedback for expedited resubmission.	**Working**

MODULE 12: NOTIFICATION MODULE

Sub-Module 12.1: Automated Notifications

Specific Objective: This function triggers in-app and email alerts to notify authorized users of related updates.

Table 12.1A. Functional Test result for Automated Notifications

Automated Notifications - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Automated Notifications	Verify triggering of in-app and email alerts to notify authorized users of related updates (new reports, comments, account changes).	Notification mailer and database notification channel configured.	1. Administrator approves a student report. 2. Check Guardian in-app notification bell icon. 3. Check simulated/logged outbound email queue. 4. Post a comment on a report and check Teacher in-app notification.	In-app badge count increments immediately; notification record created in database; email alert queued/dispatched.	**Pass**

Table 12.1B. Validation Test result for Automated Notifications

Automated Notifications - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Automated Notifications	System Update Notification Dispatch	System triggers notification event upon critical report, goal, or account updates.	System dispatches in-app database alert and sends email message to authorized user.	System successfully dispatched in-app notifications and email alerts to authorized recipients.	**Working**

MODULE 13: DATABASE MODULE

Sub-Module 13.1: Data Storage

Specific Objective: Manages the centralized storage of user credentials, student profiles, IEP data, and file directory paths.

Table 13.1A. Functional Test result for Data Storage

Data Storage - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Data Storage	Verify centralized storage of user credentials, student profiles, IEP data, and file directory paths with relational integrity.	Database engine initialized and migrations executed.	1. Create user, student profile, IEP goals, and upload a document. 2. Query database tables directly (users, students, ieps, goals, documents).	All records persist with accurate foreign key references and timestamps.	**Pass**

Table 13.1B. Validation Test result for Data Storage

Data Storage - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Data Storage	Relational Data Insertion & Retrieval	System persists student records, user accounts, and document paths.	Database verifies schema integrity, primary/foreign keys, and storage paths.	System successfully stored and indexed centralized data records with complete referential integrity.	**Working**

Sub-Module 13.2: Data Security

Specific Objective: Implements encryption and compliance measures to ensure data confidentiality and integrity.

Table 13.2A. Functional Test result for Data Security

Data Security - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Data Security	Verify encryption and compliance measures ensuring data confidentiality and integrity across passwords, sensitive student records, and tokens.	Application encryption key (APP_KEY) configured in environment.	1. Register user and inspect password column in users table. 2. Store sensitive student diagnostic information and inspect database column. 3. Inspect session tokens and file vault access tokens.	Passwords hashed using bcrypt/argon2; sensitive medical/diagnostic fields stored encrypted; session tokens cryptographically signed.	**Pass**

Table 13.2B. Validation Test result for Data Security

Data Security - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Data Security	Data Encryption & Decryption Cycle	System encrypts sensitive student data columns and checks password hashing.	Database stores encrypted ciphertext and decrypts exclusively in authorized application context.	System successfully enforced cryptographic encryption on sensitive records and hashed credentials.	**Working**

Sub-Module 13.3: Scalability

Specific Objective: Ensures the database architecture can seamlessly accommodate continuous data growth.

Table 13.3A. Functional Test result for Scalability

Scalability - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Scalability	Verify database architecture seamlessly accommodates continuous data growth through indexed lookups, pagination, and optimized queries.	High data volume populated (students, daily logs, reports).	1. Load report lists, student directories, and daily log indexes containing extensive record sets. 2. Measure query response times and inspect SQL queries for eager loading (with()).	Queries execute under indexed keys; pagination limits memory usage; response times remain consistently rapid.	**Pass**

Table 13.3B. Validation Test result for Scalability

Scalability - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Scalability	High-Volume Query Execution & Pagination	Application loads large dataset collections across academic reporting modules.	Database utilizes indexed fields and paginated limits to maintain low query latency.	System successfully maintained fast response times and low memory consumption under high data volume.	**Working**

Sub-Module 13.4: Backup and Recovery

Specific Objective: Executes regular backups and recovery protocols to prevent data loss.

Table 13.4A. Functional Test result for Backup and Recovery

Backup and Recovery - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Backup and Recovery	Verify execution of regular backups and recovery protocols to prevent data loss.	Backup configuration active and database write access enabled.	1. Execute database backup routine (php artisan db:backup or snapshot dump). 2. Verify generated backup archive integrity and checksum. 3. Simulate recovery protocol on test database from backup file.	Backup file generated successfully; recovery restores complete dataset without data corruption or loss.	**Pass**

Table 13.4B. Validation Test result for Backup and Recovery

Backup and Recovery - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Backup and Recovery	Database Backup Generation & Restoration	Administrator triggers automated database backup snapshot and simulates restoration.	System creates intact SQL/archive file and recovers database schema without data corruption.	System successfully generated verifiable database backup snapshots and executed recovery protocol.	**Working**

MODULE 14: STUDENT LIST MODULE

Sub-Module 14.1: Assigned Learner Directory

Specific Objective: This navigation feature provides Teachers with a dedicated interface displaying a list of their currently assigned students, allowing them to instantly view profiles or print finalized IEP documents.

Table 14.1A. Functional Test result for Assigned Learner Directory

Assigned Learner Directory - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Assigned Learner Directory	Verify dedicated navigation interface displaying list of currently assigned students for Teachers, allowing instant profile view and finalized IEP document printing.	Teacher is authenticated; students are assigned to teacher's classroom.	1. Navigate to /teacher/students. 2. Inspect roster of assigned learners. 3. Click 'View Profile' on a student card. 4. Click 'Print Finalized IEP' button.	Directory renders assigned students only; profile opens instantly; finalized IEP prints/streams formatted document.	**Pass**

Table 14.1B. Validation Test result for Assigned Learner Directory

Assigned Learner Directory - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Assigned Learner Directory	Assigned Learner Directory Access & Printing	Teacher navigates assigned student directory, views profile, and prints finalized IEP.	System scopes learner query to assigned teacher and compiles printable IEP document.	System successfully displayed assigned learner directory and enabled instant profile viewing and IEP printing.	**Working**

MODULE 15: ACADEMIC CALENDAR CONFIGURATION MODULE

Sub-Module 15.1: Period Definition

Specific Objective: This function allows administrators to configure the official start and end dates for Quarterly periods, establishing the timeframe for AI summary generation and system archiving, as well as initializing the new school year.

Table 15.1A. Functional Test result for Period Definition

Period Definition - Functional Testing					
Feature	**Test Case Description**	**Precondition**	**Test Steps**	**Expected Results**	**Pass/Fail**
Period Definition	Verify Administrator configuration of official start and end dates for Quarterly periods, establishing timeframe for AI summary generation, system archiving, and initializing new school year.	Administrator is authenticated.	1. Navigate to /admin/calendar (or /admin/quarters). 2. Click 'Initialize New School Year' and specify school year (e.g., '2026-2027'). 3. Define start and end dates for Q1, Q2, Q3, and Q4. 4. Save period configurations.	Quarterly periods persist in database; date boundaries establish timeframes for AI report summaries and archiving triggers.	**Pass**

Table 15.1B. Validation Test result for Period Definition

Period Definition - Validation Testing					
Function to Test	**Event**	**Test Case**	**Attempt to**	**Result**	**Remarks**
Period Definition	Quarterly Period Date Configuration	Administrator inputs quarterly date boundaries and initializes new academic year.	System validates chronological continuity of dates and establishes operational academic periods.	System successfully configured quarterly period boundaries, establishing timeframes for AI summaries and archiving.	**Working**